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Cambridge IGCSE Combined and Co-ordinated Sciences

C2 Atoms, Elements and Compounds

Pages 273-300

Comprehensive Exam Preparation Guide

C2 — Atoms, Elements and Compounds

1. Elements, Compounds and Mixtures (C2.01)

Elements

Every substance is made up of **atoms** — the incredibly small particles from which the whole material world is built.

- **Element** — a substance made up of only **one type of atom**; it cannot be broken down into anything simpler by chemical reactions.
- All the atoms of an element contain the **same number of protons**.
- There are **118** known elements, and **94** of them are found naturally on Earth.
- Most of the known mass of the universe is just two elements: **hydrogen (92%)** and **helium (7%)**. All the other elements make up only 1%.
- Just **eight** elements make up more than 98% of the mass of the Earth's crust. **Silicon** and **oxygen**, bound together in silicate rocks, make up almost three-quarters of it.

Metals and Non-metals

Elements can be divided into **metals** and **non-metals**. Their different properties come from their different structure and bonding (see sections 5 to 8).

Property	Metals	Non-metals
Position in the Periodic Table	Left and centre	Top right
Electrical conductivity	Good conductors	Poor conductors (graphite is an exception)
When a force is applied	Malleable: can be bent and beaten into sheets	Brittle if solid
Ions formed	Lose outer electrons to form positive ions, e.g. Na^+, Mg^{2+}	Gain electrons to form negative ions, e.g. Cl^-, O^{2-}
Bonding in the element	Metallic bonding	Covalent bonding (the noble gases exist as single atoms)

Compounds and Mixtures

- **Compound** — two or more elements **chemically combined** in **fixed proportions**; the ratio of the different types of atom is always the same.
- **Mixture** — two or more elements or compounds mixed together **without being chemically bonded**; the amounts of each can vary.

Example — Iron and Sulfur

- A **mixture** of iron and sulfur powders can contain any amounts of the two elements. Each element keeps its own properties, so the iron can be separated out with a **magnet**.
- When the mixture is **heated**, the elements react to form a **black compound, iron sulfide**, which is **not attracted to a magnet**.

iron + sulfur → iron sulfide

Comparing Compounds and Mixtures

Compounds	Mixtures
A compound is a single substance	A mixture contains two or more substances
The composition is always the same	The composition can be varied
Formation involves a chemical reaction	No chemical change takes place when a mixture is made
The properties are very different from those of the elements in the compound	The properties of the substances in the mixture are still present
Can only be broken down by chemical reactions	The substances can be separated by physical methods

2. Atomic Structure and the Periodic Table (C2.02)

The Structure of the Atom

- Modern ideas about atoms are based on the **atomic theory** put forward by **John Dalton in 1807**: atoms are the basic building blocks of the elements, and they can join together to make molecules.
- Later research showed that atoms are made of even smaller **subatomic particles**. The **electron** was discovered in **1897**, soon followed by the **proton**. By **1932**, when the **neutron** was discovered, it was clear that all atoms are made of protons, neutrons and electrons.
- At the centre of the atom is a very small, positively charged **nucleus**, which contains the protons and neutrons and almost **all the mass** of the atom.
- The negatively charged **electrons** are arranged in **shells** around the nucleus. Most of the atom is the space occupied by the electrons.
- The **atom** is the smallest particle that shows the chemical characteristics of a particular element.

Protons, Neutrons and Electrons

Subatomic particle	Relative mass	Relative charge	Location in atom
Proton	1	+1	In the nucleus
Neutron	1	0	In the nucleus
Electron	1/1840 (negligible)	−1	Outside the nucleus, in shells

- Protons and neutrons have almost the **same mass**. Electrons have virtually **no mass** (1/1840, or 0.00054, of the mass of a proton).
- The electrons are held in the atom by the **electrostatic force of attraction** between them and the positive protons in the nucleus.
- An atom is **electrically neutral**: it has **equal numbers of protons and electrons**, so the positive charge of the nucleus is balanced by the negative charge of the electrons.
- **Hydrogen** is the only atom with **no neutrons**: it has 1 proton and 1 electron.
- **Helium** has 2 protons, 2 neutrons and 2 electrons. Its charges balance: (+2) + (−2) = 0. Its mass is 2 + 2 = 4 units, because the electrons add almost nothing.
- The number of neutrons needed to hold the nucleus together **increases** as atoms get larger. A gold atom has 79 protons, 118 neutrons and 79 electrons.

Proton Number and Mass Number

- **Proton number (atomic number), Z** — the number of **protons** in the nucleus of an atom. It identifies the element: only hydrogen atoms have 1 proton, only helium atoms have 2 and only gold

atoms have 79.

- **Mass number (nucleon number), A** — the total number of **protons and neutrons** in the nucleus. Protons and neutrons together are called **nucleons**.
- An atom is written with the mass number at the **top left** of the symbol and the proton number at the **bottom left**: $^{4}_{2}\mathrm{He}$, $^{7}_{3}\mathrm{Li}$, $^{12}_{6}\mathrm{C}$, $^{16}_{8}\mathrm{O}$, $^{238}_{92}\mathrm{U}$.

To work out the particles in an atom:

- Number of protons = proton number (Z)
- Number of electrons = number of protons (Z), because the atom is neutral
- Number of neutrons = mass number − proton number (A − Z)

Atom	Symbol	Z	A	Protons	Neutrons (A − Z)	Electrons
Hydrogen	$^{1}_{1}\mathrm{H}$	1	1	1	0	1
Helium	$^{4}_{2}\mathrm{He}$	2	4	2	2	2
Lithium	$^{7}_{3}\mathrm{Li}$	3	7	3	4	3
Carbon	$^{12}_{6}\mathrm{C}$	6	12	6	6	6
Oxygen	$^{16}_{8}\mathrm{O}$	8	16	8	8	8
Calcium	$^{40}_{20}\mathrm{Ca}$	20	40	20	20	20
Gold	$^{197}_{79}\mathrm{Au}$	79	197	79	118	79
Uranium	$^{238}_{92}\mathrm{U}$	92	238	92	146	92

Worked Example — Particles in $^{23}_{11}\mathrm{Na}$

1. The proton number is 11, so there are **11 protons**.
2. The atom is neutral, so there are also **11 electrons**.
3. The mass number is 23, so the number of neutrons is 23 − 11 = **12 neutrons**.

Electronic Configuration

In 1913 **Niels Bohr**, working with Ernest Rutherford, developed a theory of how the electrons are arranged in atoms:

- Electrons orbit the nucleus in **electron shells (energy levels)**, which have different energies.
- The shells are filled starting with the one with the **lowest energy**, **closest to the nucleus**. Shells further from the nucleus have higher energies.
- The **first shell** can hold only **2** electrons.
- The **second shell** holds up to **8** electrons. For the first 20 elements the **third shell** also holds up to **8**; in later elements it can expand to hold 18.
- **Full shells** are the most stable arrangement.

The **electronic configuration** gives the number of electrons in each shell, starting from the innermost shell, e.g. sodium is **2,8,1**.

Worked Example — Electronic Configuration of Aluminium

1. Aluminium has proton number 13, so the atom has 13 electrons.
2. The first shell takes 2 electrons, leaving 11.
3. The second shell takes 8 electrons, leaving 3.
4. The remaining 3 electrons go into the third shell, so the configuration is **2,8,3**.

- When compounds of some metals are heated in a flame, electrons in the atoms move between two different shells and light of a characteristic colour is given out. This is the basis of **flame tests**; the

aurora borealis is caused in a similar way, by radiation from the Sun moving the electrons in atoms of the gases in the atmosphere.

Electronic Configuration and the Periodic Table

- **Group number** = the number of electrons in the **outer shell**.
- **Period number** = the number of **occupied shells**.
- Example: silicon is in **Group IV, Period 3**, so it has 4 outer electrons and 3 shells: **2,8,4**.
- Working backwards: an atom with the configuration 2,8,7 has 7 outer electrons and 3 shells, so it is in Group VII, Period 3. It has 2 + 8 + 7 = 17 electrons, so Z = 17: it is **chlorine**.

The electronic configurations of the first 20 elements, set out as in the Periodic Table:

Period	Group I	Group II	Group III	Group IV	Group V	Group VI	Group VII	Group VIII
1	H 1 (placed on its own)							He 2
2	Li 2,1	Be 2,2	B 2,3	C 2,4	N 2,5	O 2,6	F 2,7	Ne 2,8
3	Na 2,8,1	Mg 2,8,2	Al 2,8,3	Si 2,8,4	P 2,8,5	S 2,8,6	Cl 2,8,7	Ar 2,8,8
4	K 2,8,8,1	Ca 2,8,8,2						

The Noble Gas Electronic Configuration

- The **noble gases** (Group VIII) are very unreactive gases.
- Their atoms have a **full outer shell** of electrons — 8 outer electrons (helium has 2) — which is a very **stable** arrangement.
- Noble gas atoms do not form bonds with other atoms by sharing or transferring electrons, so they exist as **separate atoms**.
- Atoms of other elements react by losing, gaining or sharing outer electrons so that they end up with a **noble gas electronic configuration**.

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3. Isotopes (C2.03)

What Are Isotopes?

- **Isotopes** — atoms of the same element that have the **same proton number** but a **different nucleon (mass) number**.
- The difference comes from **different numbers of neutrons** in the nucleus; the numbers of protons and electrons are the same.
- Isotopes are named using their mass number, e.g. **carbon-12**, **carbon-13** and **carbon-14**. Their symbols show both numbers, e.g. $^{12}_{6}C$ and $^{14}_{6}C$.
- Pure samples of many elements, such as hydrogen, carbon and chlorine, naturally contain a **mixture of isotopes**.

Isotope (natural abundance)	Symbol	Protons	Neutrons	Electrons
Hydrogen (99.99%)	$^{1}_{1}H$	1	0	1
Deuterium (0.01%)	$^{2}_{1}H$	1	1	1
Tritium (made artificially)	$^{3}_{1}H$	1	2	1
Carbon-12 (98.9%)	$^{12}_{6}C$	6	6	6

Isotope (natural abundance)	Symbol	Protons	Neutrons	Electrons
Carbon-13 (1.1%)	$^{13}_{6}C$	6	7	6
Carbon-14 (trace)	$^{14}_{6}C$	6	8	6
Neon-20 (90.5%)	$^{20}_{10}Ne$	10	10	10
Neon-21 (0.3%)	$^{21}_{10}Ne$	10	11	10
Neon-22 (9.2%)	$^{22}_{10}Ne$	10	12	10
Chlorine-35 (75%)	$^{35}_{17}Cl$	17	18	17
Chlorine-37 (25%)	$^{37}_{17}Cl$	17	20	17

- **Tritium** and **carbon-14** are **radioactive isotopes** because their nuclei are **unstable**.

Properties of Isotopes

- Isotopes of an element have the **same chemical properties** because they have the **same number of electrons**, and so the **same electronic configuration**. It is the number of electrons in an atom that decides how it forms bonds and reacts with other atoms.
- Some **physical properties** of isotopes are different, e.g. their mass, and some isotopes are radioactive.

Symbols for Atoms and Ions

The same rules work for ions. A positive ion has **lost** electrons and a negative ion has **gained** electrons; the protons and neutrons do not change.

Particle	Protons	Neutrons	Electrons
$^{35}_{17}Cl$ atom	17	35 − 17 = 18	17
$^{35}_{17}Cl^-$ ion	17	18	17 + 1 = 18
$^{40}_{20}Ca^{2+}$ ion	20	40 − 20 = 20	20 − 2 = 18
$^{16}_{8}O^{2-}$ ion	8	16 − 8 = 8	8 + 2 = 10

4. Chemical Bonding (C2.04)

Elements Combine to Make Compounds

- Each compound is made of particular elements, always combined in the **same proportions**. For example, each **water** molecule contains **two hydrogen atoms** bonded to **one oxygen atom**.
- When elements react to form a compound, their characteristic properties are **replaced** by those of the compound. The properties of a compound bear no direct relationship to those of the elements that make it up.

Substance	Properties
Sodium	A highly reactive metal; fizzes violently with water and may even cause a flame
Chlorine	A toxic gas, once used as a chemical weapon and still used to kill harmful microbes in water treatment
Sodium chloride (common salt)	A harmless crystalline solid, used to flavour food and essential for life

Burning sodium in chlorine gas produces a white smoke of sodium chloride:

sodium + chlorine → sodium chloride

$2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$

Chemical Formulae

- A **chemical formula** is a shorthand way of representing elements and compounds using the **symbols** of the elements.
- The formula shows the **ratio of atoms** in the compound: **NaCl** has 1 sodium to 1 chlorine, and **H_2O** has 2 hydrogen atoms to 1 oxygen atom.

Types of Chemical Bonding

- **Chemical bonding** — the strong forces that hold atoms (or ions) together in a substance. Whatever the type of bonding, it is always the **outer electrons** of the atoms that are involved.
- **Molecular compounds** — atoms bonded together by **shared electrons**, e.g. water, ammonia and methane.
- **Ionic compounds** — many **ions** (charged particles) held together in a regular structure by the attraction between their opposite charges, e.g. sodium chloride.

Type of substance	Type of bonding	Structure formed	Examples
Metal elements	Metallic	Giant metallic lattice	Copper
Metal + non-metal(s)	Ionic: electrons transferred	Giant ionic lattice	NaCl, MgO
Non-metal elements, and non-metal + non-metal(s)	Covalent: electrons shared	Simple molecules or giant molecular lattices	Cl_2, H_2O, CH_4; diamond, SiO_2
Noble gases	No bonding	Separate atoms	He, Ne, Ar

5. Ions and Ionic Bonds (C2.05)

Formation of Ions

- **Ion** — a charged particle formed when an atom (or a group of atoms) **loses or gains electrons**.
- Most atoms do not have a full outer shell. One way to gain the stable **noble gas configuration** is to **transfer** electrons completely from one atom to another.
- **Cation** — a **positive ion**, formed when a **metal atom loses** its outer electrons. The charge equals the number of electrons lost.
- **Anion** — a **negative ion**, formed when a **non-metal atom gains** electrons. The charge equals the number of electrons gained.

A sodium atom (Group I) loses its one outer electron:

Na (2,8,1) → Na^+ (2,8) + e^-

The sodium ion has a **single positive charge** because it has only 10 electrons but still has 11 protons in its nucleus. It now has a complete outer shell, the stable configuration of **neon**.

A chlorine atom (Group VII) gains one electron:

Cl (2,8,7) + e^- → Cl^- (2,8,8)

The chloride ion has a **single negative charge** because it has one more electron (18) than there are protons in its nucleus (17). It now has the configuration of **argon**.

Ionic Bonding

Ionic bonding occurs in compounds of **metals with non-metals**. Electrons are **transferred** from the metal atoms to the non-metal atoms, and the oppositely charged ions formed are held together by **strong electrostatic forces of attraction**. This is the **ionic bond**.

The important features of ionic bonding are:

- The electrons involved in forming the ions are those in the **outer shell** of the atoms.
- **Metal atoms lose** their outer electrons to become **positive ions**. The charge equals the number of electrons lost.
- Generally, **non-metal atoms gain** electrons to become **negative ions**. The charge equals the number of electrons gained.
- The ions formed have the stable electron arrangement of the **nearest noble gas** in the Periodic Table.

Dot-and-Cross Diagrams for Ionic Compounds

In a dot-and-cross diagram, the electrons of one atom are drawn as **dots** and those of the other atom as **crosses**, usually showing the **outer electrons only**. An arrow shows the electron transfer, and each ion formed is drawn in **square brackets** with its **charge** at the top right.

Compound	Electron transfer	Ions formed	Formula
Sodium chloride	Na (2,8,1) gives its 1 outer electron to Cl (2,8,7)	Na^+ (2,8) and Cl^- (2,8,8)	NaCl
Magnesium oxide	Mg (2,8,2) gives its 2 outer electrons to O (2,6)	Mg^{2+} (2,8) and O^{2-} (2,8)	MgO
Calcium chloride	Ca (2,8,8,2) gives 1 electron to each of two Cl atoms (2,8,7)	Ca^{2+} (2,8,8) and two Cl^- (2,8,8)	$CaCl_2$

- In the drawing of the ions, the metal ion has an **empty outer shell** because its outer electrons have gone. The non-metal ion has **8 outer electrons**: its own dots plus the transferred cross or crosses.
- In magnesium oxide, both ions have the configuration of **neon**; in calcium chloride, all the ions have the configuration of **argon**.
- The **formula** follows from the charges, because the total positive charge must equal the total negative charge: Mg^{2+} and O^{2-} give **MgO**, while Ca^{2+} needs two Cl^- ions, giving **$CaCl_2$**.

Physical Properties of Ionic Compounds

In the solid, the ions are arranged in a **giant ionic lattice**: a regular arrangement in which each ion is surrounded by ions of the **opposite charge**. The whole structure is held together by the electrostatic forces of attraction between the oppositely charged ions.

Property of ionic compounds	Reason
Solids at room temperature	There is a regular arrangement of the ions in a lattice; ions with opposite charges are next to each other
High melting and boiling points	The ions are attracted to each other by strong electrostatic forces; large amounts of energy are needed to separate them
Often soluble in water (not usually soluble in organic solvents such as ethanol or methylbenzene)	Water is attracted to charged ions, so many ionic solids dissolve in water
Conduct electricity when molten or dissolved in water, but not when solid	In the liquid or solution the ions are free to move towards the electrodes when a voltage is applied; in the

Property of ionic compounds	Reason
	solid they are held in fixed positions

6. Simple Molecules and Covalent Bonds (C2.06)

Covalent Bonding

- Most non-metal elements are made of **molecules**, not separate atoms; only the **noble gases** exist as individual atoms. Hydrogen (H_2), oxygen (O_2), nitrogen (N_2) and chlorine (Cl_2) consist of **diatomic molecules**.
- **Covalent bonding** — bonding formed by the **sharing of one or more pairs of electrons** between atoms. It is the bonding in **non-metal elements** and in compounds of **one non-metal with another**.
- A covalent bond is held together by the attraction between the **shared pair of electrons** (negative) and the **nuclei** (positive) of the two atoms.

The main features of covalent bonding are:

- The bond is formed by the **sharing of a pair of electrons** between two atoms, leading to **noble gas electronic configurations**.
- Generally, each atom contributes an **equal number** of electrons to each bond (one electron each for a single bond).

Other key points:

- **Hydrogen** atoms gain a share of **2** electrons (like helium); atoms of other elements gain a share of **8** outer electrons.
- A **single bond** is one shared pair (drawn as a line, –), a **double bond** is two shared pairs (=) and a **triple bond** is three shared pairs (≡). These lines are used in **displayed formulae**.
- **Dot-and-cross diagrams** show the outer electrons only, because the inner electrons are not involved in the bonding.
- The number of covalent bonds an atom usually forms equals the number of electrons it needs to complete its outer shell: **H 1, Cl 1, O 2, N 3, C 4**.

Dot-and-Cross Diagrams for Simple Molecules

Molecule	Atoms used	Bonds (displayed formula)	What the dot-and-cross diagram shows
Hydrogen, H_2	2 H (1)	One single bond, H–H	One shared pair (one dot, one cross); each H has a share of 2 electrons
Chlorine, Cl_2	2 Cl (2,8,7)	One single bond, Cl–Cl	One shared pair; each Cl also has 3 non-bonding pairs, giving it 8 outer electrons (each Cl is now 2,8,8)
Hydrogen chloride, HCl	H (1) + Cl (2,8,7)	One single bond, H–Cl	One shared pair; Cl also has 3 non-bonding pairs
Water, H_2O	2 H (1) + O (2,6)	Two single bonds, H–O–H	Two shared pairs; O also has 2 non-bonding pairs
Ammonia, NH_3	3 H (1) + N (2,5)	Three single N–H bonds	Three shared pairs; N also has 1 non-bonding pair
Methane, CH_4	4 H (1) + C (2,4)	Four single C–H bonds	Four shared pairs; C has no non-bonding pairs

Molecule	Atoms used	Bonds (displayed formula)	What the dot-and-cross diagram shows
Oxygen, O_2	2 O (2,6)	One double bond, O=O	Two shared pairs (each O gives 2 electrons); each O also has 2 non-bonding pairs
Nitrogen, N_2	2 N (2,5)	One triple bond, N≡N	Three shared pairs (each N gives 3 electrons); each N also has 1 non-bonding pair
Carbon dioxide, CO_2	C (2,4) + 2 O (2,6)	Two double bonds, O=C=O	C shares 2 electrons with each O; each O also has 2 non-bonding pairs
Ethene, C_2H_4	2 C (2,4) + 4 H (1)	One C=C double bond and four C–H single bonds	Each C shares 2 electrons with the other C and 1 electron with each of its two H atoms
Methanol, CH_3OH	C (2,4) + O (2,6) + 4 H (1)	Three C–H, one C–O and one O–H, all single bonds	O also has 2 non-bonding pairs

Physical Properties of Simple Covalent Compounds

There are **two levels of force** in a simple molecular substance:

- The **covalent bonds within** each molecule are **strong** and difficult to break.
- The **intermolecular forces between** the molecules are only **weak** and are easily broken.

When a simple molecular substance melts or boils, only the weak intermolecular forces are overcome; the covalent bonds do not break. So only a small amount of energy is needed to let the molecules move away from each other.

Property of simple covalent compounds	Reason
Often liquids or gases at room temperature	They are made of simple molecules; the atoms are joined together by covalent bonds
Low melting and boiling points	The forces between the molecules (intermolecular forces) are only very weak; not much energy is needed to move the molecules further apart
Do not conduct electricity	There are no ions present to carry the current

7. Giant Covalent Structures (C2.07)

Giant covalent structures (macromolecules, or giant molecular crystals) contain very many atoms held together by **strong covalent bonds** throughout the whole structure. Examples are the element carbon, as **diamond** and **graphite**, and compounds such as **silica, SiO_2**.

Diamond

- Each carbon atom is covalently bonded to **four** others, arranged **tetrahedrally**. The strong bonds extend in all directions through the whole crystal.
- **Diamond has a very high melting point** — a lot of energy is needed to break the many strong covalent bonds.
- **Diamond is very hard** — it is the hardest natural substance, so it is used in **cutting tools**. The bonds are rigid, so diamond is much more **brittle** than a metal.

- **Diamond does not conduct electricity** — all the outer electrons are used to form covalent bonds, so there are **no electrons free to move**. Diamond is a typical non-metallic element.

Graphite

- The carbon atoms are arranged in **flat layers of linked hexagons**. Each layer is a two-dimensional giant molecule.
- Within each layer, every carbon atom is bonded to **three** others by strong covalent bonds, so graphite also has a **very high melting point**.
- **Graphite is soft and slippery** — between the layers there are only **weaker forces of attraction**, so the layers can **slide over each other** easily. It is used as a **lubricant** and in **pencils**, where thin layers of graphite are left on the paper.
- **Graphite conducts electricity** — each carbon atom uses only three of its four outer electrons in covalent bonds, and the **free electrons** can move between the layers carrying charge, in a similar way to metals.

Comparing Diamond and Graphite

Feature	Diamond	Graphite
Bonds per carbon atom	4, arranged tetrahedrally in three dimensions	3, in flat layers of hexagons
Appearance	Colourless, transparent crystals that sparkle in light	Dark grey, shiny solid
Hardness	The hardest natural substance	Soft: the layers can slide over each other and the solid has a slippery feel
Electrical conductivity	Does not conduct electricity	Conducts electricity
Uses	In jewellery and ornamental objects; in drill bits, diamond saws and glasscutters	In pencils and as a lubricant; as electrodes and for the brushes in electric motors

8. Metallic Bonding (C2.08)

The Structure of Metals

- Metal atoms have relatively **few electrons in their outer shells**. When they are packed together, each metal atom loses its outer electrons into a **'sea' of delocalised electrons**.
- Having lost electrons, the atoms become **positive ions** (the number of protons in the nucleus is unchanged). A metal is therefore made of **positive ions** packed together and surrounded by electrons that can **move freely** between the ions.
- The delocalised electrons are not restricted to orbiting one positive ion. They act as a kind of electrostatic 'glue' that holds the structure together.
- **Metallic bonding** — the **electrostatic force of attraction** between the mobile 'sea' of electrons and the regular array of **positive metal ions**.

Physical Properties of Metals

Property	Explanation using metallic bonding
Good conductors of electricity	The mobile electrons can move through the structure, carrying the current; alloys such as solder and brass also conduct for the same reason

Property	Explanation using metallic bonding
Malleable: easily bent and shaped	The positive ions are arranged in layers. When a force is applied, the layers can slide over each other. The attractive forces act in all directions, so new bonds are easily formed and the metal is left with a new shape instead of breaking

Summary of Structure and Bonding

Structure	Particles	Bonding	Typical properties	Examples
Giant ionic lattice	Positive and negative ions	Ionic: strong electrostatic attraction between oppositely charged ions	High melting and boiling points; often soluble in water; conduct only when molten or dissolved	NaCl, MgO, KCl
Simple molecules	Small molecules	Strong covalent bonds within molecules; weak intermolecular forces between them	Low melting and boiling points; often liquids or gases; do not conduct	H_2O, CH_4, Cl_2, I_2
Giant covalent structure	Atoms	Strong covalent bonds throughout	Very high melting points; do not conduct, except graphite	Diamond, graphite, SiO_2
Giant metallic lattice	Positive metal ions in a sea of delocalised electrons	Metallic bonding	Conduct electricity; malleable	Copper, brass

9. Key Terms

Term	Definition
Atom	The smallest particle of an element that shows the chemical characteristics of that element
Element	A substance that cannot be further divided into simpler substances by chemical methods; all the atoms of an element contain the same number of protons
Compound	A substance formed by the chemical combination of two or more elements in fixed proportions
Mixture	Two or more elements or molecules mixed together without being chemically bonded
Molecule	A particle made of two or more atoms joined together by covalent bonds, e.g. H_2, H_2O
Subatomic particles	Very small particles — protons, neutrons and electrons — from which all atoms are made
Proton	A subatomic particle with a relative atomic mass of 1 and a charge of +1 found in the nucleus of all atoms

Term	Definition
Neutron	An uncharged subatomic particle present in the nucleus of atoms; a neutron has a mass of 1 relative to a proton
Electron	A subatomic particle with negligible mass and a relative charge of −1; electrons are present in all atoms, located in the shells (energy levels) outside the nucleus
Nucleus	The central region of an atom that is made up of the protons and neutrons of the atom; the electrons orbit around the nucleus in different shells or energy levels
Nucleons	The particles in the nucleus of an atom: protons and neutrons
Proton number (atomic number), Z	The number of protons in the nucleus of an atom
Mass number (nucleon number), A	The total number of protons and neutrons in the nucleus of an atom
Atomic theory	A model of the atom in which electrons can only occupy certain shells (or energy levels) moving outwards from the nucleus of an atom
Electron shells (energy levels)	The allowed energies of electrons in atoms; electrons fill these shells (or levels) starting with the one closest to the nucleus
Electronic configuration	A shorthand method of describing the arrangement of electrons within the electron shells (energy levels) of an atom, also called the electronic structure; it is given as the number of electrons in the first shell, then the second shell, then the third shell and so on, e.g. 2,8,1
Group number	The number of the vertical column an element is in on the Periodic Table; it equals the number of electrons in the outer shell
Period number	The number of the horizontal row an element is in on the Periodic Table; it equals the number of occupied electron shells
Noble gases	Elements in Group VIII; a group of stable, very unreactive gases whose atoms have full outer shells
Isotopes	Atoms of the same element that have the same proton number but a different nucleon number, because they have different numbers of neutrons
Radioactive isotope	An isotope with an unstable nucleus, e.g. tritium and carbon-14
Chemical formula	A shorthand method of representing chemical elements and compounds using the symbols of the elements
Chemical bonding	The strong forces that hold atoms (or ions) together in the various structures that chemical substances can form; this includes metallic bonding, covalent bonding and ionic bonding
Ions	Charged particles made from an atom, or groups of atoms (compound ions), by the loss or gain of electrons
Cation	A positive ion, formed when an atom loses electrons, e.g. Na^{+}, Mg^{2+}
Anion	A negative ion, formed when an atom gains electrons, e.g. Cl^{-}, O^{2-}

Term	Definition
Ionic bonding	A strong electrostatic force of attraction between oppositely charged ions
Electrostatic forces	Strong forces of attraction between particles with opposite charges; such forces are involved in ionic bonding
Giant ionic lattice	A regular arrangement of alternating positive and negative ions, in which each ion is surrounded by ions of the opposite charge
Covalent bonding	Chemical bonding formed by the sharing of one or more pairs of electrons between atoms
Diatomic molecules	Molecules containing two atoms, e.g. hydrogen, H_2
Single, double and triple bonds	Covalent bonds made of one, two or three shared pairs of electrons, e.g. Cl–Cl, O=O, N≡N
Dot-and-cross diagram	A diagram drawn to represent the bonding in a molecule, or the electrons in an ion; usually only the outer electrons are shown, and they are represented by dots or crosses depending on which atom they are from
Displayed formula	A formula that shows every atom in a molecule and every covalent bond as a line, e.g. H–Cl, O=C=O
Intermolecular forces	The weak forces of attraction between molecules
Hydrocarbon	A compound of carbon and hydrogen only, e.g. methane, CH_4
Giant covalent structure	A giant structure (macromolecule) in which very many atoms are held together by strong covalent bonds, e.g. diamond, graphite and silica, SiO_2
'Sea' of delocalised electrons	The free, mobile electrons between the positive ions in a metallic lattice
Metallic bonding	An electrostatic force of attraction between the mobile 'sea' of electrons and the regular array of positive metal ions within a solid metal
Malleable	Describes the property that allows metals to be bent and beaten into sheets

10. Key Equations

Counting Subatomic Particles

1. Number of protons = proton number (Z)
2. Number of electrons in a neutral atom = number of protons = Z
3. Mass number (A) = number of protons + number of neutrons
4. Number of neutrons = A − Z
5. Electrons in a positive ion = Z − charge, e.g. Na^+ has 11 − 1 = 10 electrons
6. Electrons in a negative ion = Z + charge, e.g. Cl^- has 17 + 1 = 18 electrons

Electronic Configuration and the Periodic Table

1. Maximum number of electrons (first 20 elements): first shell 2, second shell 8, third shell 8
2. Group number = number of electrons in the outer shell
3. Period number = number of occupied shells

4. Charge on a metal ion = number of outer electrons lost (Group I gives 1+, Group II gives 2+)
5. Charge on a non-metal ion = number of electrons gained to fill the outer shell (Group VI gives 2−, Group VII gives 1−)

Forming Ions by Electron Transfer

1. Na (2,8,1) → Na^+ (2,8) + e^-
2. K (2,8,8,1) → K^+ (2,8,8) + e^-
3. Mg (2,8,2) → Mg^{2+} (2,8) + $2e^-$
4. Ca (2,8,8,2) → Ca^{2+} (2,8,8) + $2e^-$
5. Cl (2,8,7) + e^- → Cl^- (2,8,8)
6. O (2,6) + $2e^-$ → O^{2-} (2,8)

Formulae of Ionic Compounds (Charges Must Balance)

1. $Na^+ + Cl^- \rightarrow NaCl$
2. $K^+ + Cl^- \rightarrow KCl$
3. $Li^+ + F^- \rightarrow LiF$
4. $Na^+ + Br^- \rightarrow NaBr$
5. $Mg^{2+} + O^{2-} \rightarrow MgO$
6. $Ca^{2+} + 2Cl^- \rightarrow CaCl_2$

Reactions in This Chapter

1. iron + sulfur → iron sulfide
2. $Fe(s) + S(s) \rightarrow FeS(s)$
3. sodium + chlorine → sodium chloride
4. $2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$
5. potassium + chlorine → potassium chloride
6. $2K(s) + Cl_2(g) \rightarrow 2KCl(s)$
7. magnesium + oxygen → magnesium oxide
8. $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$
9. carbon + oxygen → carbon dioxide
10. $C(s) + O_2(g) \rightarrow CO_2(g)$

Covalent Bonds

1. Electrons in one covalent bond = 2 (one shared pair)
2. Electrons involved in bonding = 2 × number of covalent bonds, e.g. CH_4: 4 × 2 = 8
3. Bonds usually formed: H = 1, Cl = 1, O = 2, N = 3, C = 4

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11. Practice Questions with Full Answers

Before You Start — Question 1

Matter is made up of very small particles such as atoms or molecules, and the chemical formula for water is H_2O. From your previous studies, discuss the following questions.

(a) What do you understand that the formula of water means?

(b) What do the letters 'H' and 'O' mean in the formula?

(c) What is the difference between an atom and a molecule?

Answer:

(a) H_2O means that each **water molecule** is made of **two hydrogen atoms** chemically bonded to **one oxygen atom**. The small 2 after the H shows there are two hydrogen atoms; no number after the O means there is one oxygen atom.

(b) 'H' is the chemical symbol for the element **hydrogen** and 'O' is the symbol for the element **oxygen**.

(c) An **atom** is the smallest particle of an element that shows the chemical characteristics of that element, e.g. a single hydrogen atom, H. A **molecule** is a particle made of two or more atoms **chemically bonded** together, e.g. H_2, O_2 or H_2O. A molecule of an element contains only one type of atom (O_2); a molecule of a compound contains different types of atom (H_2O).

Before You Start — Question 2

Are you aware of there being any particles that are smaller than an atom?

Answer: Yes. Atoms are made of even smaller **subatomic particles**: **protons** and **neutrons**, found in the tiny **nucleus** at the centre of the atom, and **electrons**, which are arranged in shells around the nucleus. Protons and neutrons each have a relative mass of 1, while an electron has a relative mass of only 1/1840. Protons have a charge of +1, electrons have a charge of −1 and neutrons are uncharged.

Before You Start — Extension

Can you use a molecular model-building kit or simple construction materials such as plasticine and drinking straws to build models of two or three simple molecules that you have encountered in your earlier science courses? It would be useful to build models of compounds such as water, ammonia and methane. Use the models to discuss what you remember of the type of bonding in the molecules and how the bonds are made.

Answer: Use a ball (or a lump of plasticine) for each atom and a straw for each bond:

- **Water, H_2O** — one oxygen atom joined to two hydrogen atoms by two single bonds; the two hydrogen atoms are at an angle to each other, not in a straight line.
- **Ammonia, NH_3** — one nitrogen atom joined to three hydrogen atoms by three single bonds, forming a low pyramid with the nitrogen at the top.
- **Methane, CH_4** — one carbon atom joined to four hydrogen atoms by four single bonds that point to the corners of a tetrahedron.

All three are **covalent** molecules. Each straw represents a **shared pair of electrons**, one electron from each atom. By sharing, each hydrogen atom gains a share of 2 electrons (like helium), and the oxygen, nitrogen and carbon atoms each gain a share of 8 outer electrons — a stable noble gas configuration. That is why oxygen forms 2 bonds, nitrogen 3 and carbon 4, while each hydrogen forms only 1.

Science in Context C2.01 — Question 1

Scientists can now see and move individual atoms using scanning tunnelling microscopes and atomic force microscopes. To celebrate the 2012 Olympic Games in London, a single molecule called olympicene was built. It was made up of five rings and was about 100 000 times thinner than a human hair. Which atoms do you think form the basic structure of olympicene? Think of the element that forms the basis of life.

Answer: Carbon atoms. Carbon is the element that forms the basis of life, and carbon atoms can bond to each other to form chains and rings. The five rings of olympicene are made of carbon atoms joined by covalent bonds, with **hydrogen** atoms attached around the outside edge (olympicene is a hydrocarbon).

Science in Context C2.01 — Question 2

Olympicene was made up of five hexagons arranged with shared sides as in a honeycomb. Predict the structure of the molecule. How many atoms do you think make up the skeleton structure of the five rings of olympicene?

Answer: The molecule is flat and made of **five hexagonal rings of carbon atoms fused together**, each ring sharing sides with its neighbours like a honeycomb. Like the Olympic rings, there are **three rings in the top row and two rings in the row below**, fitting into the gaps between them. Hydrogen atoms are bonded to the carbon atoms around the outside edge.

Counting the skeleton: five separate hexagons would need 5 × 6 = 30 carbon atoms, but an atom on a shared side belongs to more than one ring and must be counted only once. In this arrangement 11 carbon atoms belong to only one ring, 5 are shared by two rings and 3 in the middle are shared by three rings (check: 11 × 1 + 5 × 2 + 3 × 3 = 30). So the skeleton contains 11 + 5 + 3 = **19 carbon atoms**. (The molecule's formula is $C_{19}H_{12}$.)

Activity C2.01 — Timeline Task

Working in a group, investigate how our understanding of the model of the atom developed around the beginning of the 20th century, and produce a timeline. Include words and phrases such as: indivisible, subatomic particles, protons, neutrons, electrons, 'plum pudding' model, nuclear model, nucleus, orbiting electrons. Key scientists to research include J. J. Thomson, Hantaro Nagaoka, Ernest Rutherford, James Chadwick and Niels Bohr.

Answer: A model timeline:

Year	Scientist	Discovery or model
1807	John Dalton	Atomic theory: elements are made of tiny, indivisible atoms; atoms of different elements are different, and atoms join together to make molecules
1897	J. J. Thomson	Discovered the electron, a tiny negative subatomic particle, so atoms are not indivisible
1904	J. J. Thomson	'Plum pudding' model: negative electrons dotted through a sphere of positive charge
1904	Hantaro Nagaoka	'Saturnian' model: electrons orbiting a large positive centre, like the rings of Saturn
1909	Hans Geiger and Ernest Marsden, in Rutherford's laboratory	Gold foil experiment: most alpha particles passed straight through the foil, but a few bounced back
1911	Ernest Rutherford	Nuclear model: a tiny, dense, positive nucleus with orbiting electrons; the atom is mostly empty space
1913	Niels Bohr, working with Rutherford	Electrons orbit the nucleus in fixed shells (energy levels)
1917–1919	Ernest Rutherford	Showed that the nucleus contains protons, the positive particles
1932	James Chadwick	Discovered the neutron, an uncharged particle in the nucleus

Activity C2.01 — Question 1

What was remarkable about the structure of the atom suggested by the Geiger and Marsden gold foil experiments carried out in Rutherford's laboratory?

Answer: Geiger and Marsden fired positively charged alpha particles at very thin gold foil. **Most** of the particles passed **straight through**, but a **very few** were deflected through large angles and some even **bounced back**. This was remarkable because it showed that an atom is **mostly empty space**, with almost all of its mass and all of its positive charge concentrated in a **tiny, dense nucleus** at the centre,

and the electrons outside it. This **nuclear model** replaced the 'plum pudding' model, in which the positive charge was thought to be spread through the whole atom.

Activity C2.01 — Question 2

What is it about the nature of the neutron that made it the last of the particles to be discovered?

Answer: The neutron has **no electrical charge**. Protons and electrons could be detected because they are affected by electric and magnetic fields, but neutrons are not, and they pass through matter easily, so they are very hard to detect. Neutrons are also held inside the nucleus together with the protons. Their existence was suspected because atoms have more mass than their protons alone can explain, and James Chadwick finally detected them in 1932, indirectly, from their effect on other particles.

Activity C2.01 — Question 3

Once your timeline is complete, work individually to answer this question: what do you think was the most important discovery? Justify your answer.

Answer: Any discovery can be chosen if the choice is justified. A strong answer: Rutherford's discovery of the **nucleus** (1911) was the most important, because it completely changed the model of the atom. It showed that an atom is mostly empty space with a tiny, dense, positive centre, and it was the basis for Bohr's electron shells (1913) and Chadwick's neutron (1932). Other good choices are Thomson's discovery of the electron, the first evidence that atoms are made of smaller particles, or Bohr's shells, which explain bonding and the arrangement of the Periodic Table.

Question C2.01

What are the relative masses of a neutron and electron, given that a proton has a mass of 1?

Answer: Neutron: relative mass **1** (the same as a proton). Electron: relative mass **1/1840** (about 0.00054), which is negligible.

Question C2.02

How many protons, neutrons and electrons are there in an atom of phosphorus, which has a proton number of 15 and a mass number of 31?

Answer:

Protons = proton number = **15**

Electrons = number of protons, because the atom is neutral = **15**

Neutrons = mass number − proton number = 31 − 15 = **16**

Question C2.03

Explain the terms atom and element, and include a clear description of the relationship between the two terms.

Answer: An **atom** is the smallest particle of an element that still shows the chemical characteristics of that element; it has a nucleus of protons and neutrons surrounded by electrons. An **element** is a substance that cannot be broken down into simpler substances by chemical methods. The relationship: an element is made up of **only one type of atom**, because all the atoms of an element have the **same number of protons** (the same proton number). Atoms of different elements have different proton numbers, so each element has its own type of atom, with its own symbol.

Question C2.04

Explain why neutrons are important in making the nucleus of an atom stable. You need to consider the charges on the different subatomic particles present.

Answer: The nucleus contains **protons**, which are all **positively charged** and packed very close together. Like charges **repel**, so the protons push each other apart. **Neutrons** have **no charge**, so they

add to the nucleus without adding any repulsion; they separate the protons and help to hold the nucleus together. The more protons a nucleus has, the more neutrons it needs to stay stable: for example, gold has 79 protons and 118 neutrons. (Electrons are outside the nucleus, so they cannot do this job.) The only atom without neutrons is hydrogen, whose nucleus is a single proton with nothing to repel.

Experimental Skills C2.01 — Question 1

In this experiment, a nichrome wire loop is cleaned with dilute hydrochloric acid in a roaring Bunsen flame, dipped into a metal salt solution and held in the hot part of the flame, and the colour of the flame is observed. The solutions tested are, for example, NaCl(aq), KCl(aq), LiCl(aq) and $CuCl_2$(aq); $CaCl_2$(aq) and $BaCl_2$(aq) can also be used. Is it the metal or non-metal present in the salts that is responsible for the colour emitted in the flame?

Answer: The **metal**. All the salts tested are **chlorides**, so they all contain the same non-metal, yet they give **different flame colours**: lithium chloride gives red, sodium chloride yellow, potassium chloride lilac and copper(II) chloride blue-green. The hydrochloric acid used to clean the loop also contains chloride and gives no colour. So the colour must come from the **metal**: heating moves electrons in the metal atoms between shells, and when they return, light of a particular colour is given out.

Experimental Skills C2.01 — Question 2

Table C2.05 shows the general relationship between the colour, frequency and energy of the light released. The bigger the change in energy level of the electrons, the greater the energy of the light released. Which of the metal salt solutions produces light of the greatest energy? Put the salt solutions in order of the energy change involved, from the least to the greatest.

Colour	Frequency / THz
Red	405–480
Orange	480–510
Yellow	510–530
Green	530–600
Cyan (blue-green)	600–620
Blue	620–680
Violet	680–790

Energy increases down the table, from red to violet.

Answer: The flame colour shows the size of the energy change, from red (least energy) to violet (greatest). Order of energy change, from the least to the greatest:

1. Lithium chloride, LiCl — red flame (least energy)
2. Sodium chloride, NaCl — yellow flame
3. Copper(II) chloride, $CuCl_2$ — blue-green flame
4. Potassium chloride, KCl — lilac flame, at the violet end of the spectrum (greatest energy)

So **potassium chloride** produces light of the greatest energy. If they are tested too, calcium chloride (orange-red flame) comes between lithium chloride and sodium chloride, and barium chloride (light green flame) comes between sodium chloride and copper(II) chloride.

Question C2.05

(a) What are the maximum numbers of electrons that can fill the first and the second shells (energy levels) of an atom?

(b) What is the electron arrangement of a calcium atom, which has an atomic number of 20?

(c) How many electrons are there in the outer shells of the atoms of the noble gases argon and neon?

Answer:

(a) First shell: **2** electrons. Second shell: **8** electrons.

(b) Calcium has 20 electrons, arranged **2,8,8,2** (2 + 8 + 8 + 2 = 20).

(c) Both have **8** outer electrons: argon is 2,8,8 and neon is 2,8, so each has a full outer shell.

Question C2.06

By considering the subatomic particles present, explain why an atom of nitrogen is electrically neutral.

Answer: Nitrogen has proton number 7, so its nucleus contains **7 protons**, each with a charge of +1 (total +7). The atom also has **7 electrons** (arranged 2,5), each with a charge of −1 (total −7). The positive and negative charges are equal and opposite, so they **cancel out**: (+7) + (−7) = 0. The neutrons have no charge, so they do not change this. The atom therefore has no overall charge.

Question C2.07

The electronic diagrams of the atoms of four elements are shown in Figure C2.11: element A is 2,4; element B is 2,8,18,7; element C is 2,8,4; element D is 2,8,8.

(a) Which two elements are in Group IV of the Periodic Table? Explain your answer.

(b) Which element is a noble gas? Explain your answer.

(c) Which of these elements is in Group VII of the Periodic Table? Explain your answer.

(d) Which two elements are in the third period (row) of the Periodic Table? Explain your answer.

(e) What is the proton number of element C? Explain your answer.

Answer:

(a) **A and C**. Both have **4 electrons in their outer shell**, and the group number equals the number of outer electrons. (A is carbon and C is silicon.)

(b) **D**. It has a **full outer shell** of 8 electrons (2,8,8), the stable arrangement of a noble gas. (D is argon.)

(c) **B**. It has **7 electrons in its outer shell** (2,8,18,7), so it is in Group VII. (B is bromine.)

(d) **C and D**. Both have **three occupied shells**, and the period number equals the number of shells. (A has only two shells and B has four.)

(e) **14**. Element C has 2 + 8 + 4 = 14 electrons. An atom is neutral, so it has the same number of protons as electrons. (Element C is silicon.)

Reflection — Visualising Atoms and Molecules

Do you find diagrams help you to visualise the structure, movement and interaction of small objects such as atoms and molecules? If diagrams do not help, can you think of other ways that could? Discuss your ideas with a partner.

Answer: A model answer: Shell diagrams help because they show the nucleus in the centre and the electrons in their shells, so it is easy to count the particles and read off the electronic configuration; dot-and-cross diagrams later show exactly which electrons are transferred or shared. Other ways that can help are building 3D models with kits or plasticine and straws, watching animations or simulations of moving particles, and using comparisons of scale. It is important to remember that diagrams are not to scale: the nucleus is tiny compared with the whole atom.

Question C2.08

Figure C2.13 shows the nuclei of three carbon isotopes. Diagram A contains 6 protons and 6 neutrons; diagram B contains 6 protons and 8 neutrons; diagram C contains 6 protons and 7 neutrons. Which of the diagrams A, B or C represents the nucleus of an atom of the carbon-14 isotope?

Answer: B. Carbon-14 has a mass number of 14, so its nucleus contains 14 nucleons: 6 protons (as in every carbon atom) and 14 − 6 = 8 neutrons. Diagram A (6 + 6 = 12) is carbon-12 and diagram C (6 + 7 = 13) is carbon-13.

Question C2.09

What is the difference in terms of subatomic particles between an atom of chlorine-35 and an atom of chlorine-37?

Answer: Both atoms have **17 protons** and **17 electrons**. The difference is the number of **neutrons**: chlorine-35 has 35 − 17 = **18 neutrons**, while chlorine-37 has 37 − 17 = **20 neutrons**. So a chlorine-37 atom has **two more neutrons** than a chlorine-35 atom.

Question C2.10

Carbon-12 and carbon-14 are different isotopes of carbon. How many electrons are there in an atom of each isotope?

Answer: Both have **6 electrons**. Isotopes of the same element have the same proton number (6 for carbon), and a neutral atom has the same number of electrons as protons. The isotopes differ only in their number of neutrons (6 in carbon-12 and 8 in carbon-14).

Question C2.11

In a pure sample of any element, what is the most important feature of all the atoms present?

Answer: All the atoms have the **same number of protons** (the same proton number). This is what makes them all atoms of that one element. They may have different numbers of neutrons if the element has isotopes, but the proton number is always the same.

Question C2.12

A gas is made up of simple molecules that have the formula NOCl. Which of the diagrams A–D in Figure C2.27 correctly represents the molecules of NOCl? In the key, a black circle is a nitrogen atom, a white circle is a chlorine atom and a grey circle is an oxygen atom.

Diagram A: separate single atoms of nitrogen, oxygen and chlorine (four of each), not joined together.

Diagram B: molecules of three joined atoms, each made of only one type of atom (two molecules each of nitrogen, chlorine and oxygen).

Diagram C: four molecules, each made of three joined nitrogen atoms.

Diagram D: four identical molecules, each made of one nitrogen atom, one oxygen atom and one chlorine atom joined together.

Answer: D. The formula NOCl shows that each molecule contains one nitrogen atom, one oxygen atom and one chlorine atom bonded together, and a pure gas is made of identical molecules. A is a mixture of separate atoms (there are no molecules), B is a mixture of three different elements, and C is a single element containing only nitrogen.

Question C2.13

Methane is the simplest hydrocarbon (a compound of carbon and hydrogen only) and has the formula CH_4. Figure C2.28 shows its structure: a carbon atom joined to four hydrogen atoms by four single bonds. What is the total number of electrons involved in the bonding in methane?

A 10

B 2

C 8

D 4

Answer: C (8). Methane has four C–H covalent bonds, and each bond is a shared pair of 2 electrons: 4 × 2 = 8 electrons (four from the carbon atom and one from each of the four hydrogen atoms). 10 is the

total number of electrons in the molecule (6 from carbon and 4 from hydrogen), but carbon's 2 inner electrons are not involved in bonding.

Question C2.14

Covalent bonding involves electrons being shared between the atoms bonded together. Methane is made up of covalently bonded molecules. Which of the diagrams A–D in Figure C2.29 represents the bonding in methane? In the key, a dot is an electron from carbon and a cross is an electron from hydrogen.

Diagram A: a hydrogen atom in the centre surrounded by four carbon atoms, with one dot and one cross in each overlap.

Diagram B: a carbon atom (with its 2 inner electrons shown as dots) in the centre, bonded to only two hydrogen atoms, with one dot and one cross in each overlap.

Diagram C: a carbon atom (with its 2 inner electrons shown as dots) in the centre, bonded to four hydrogen atoms, with one dot and one cross in each overlap.

Diagram D: a carbon atom (with its 2 inner electrons shown as dots) in the centre, bonded to four hydrogen atoms, with two dots in each overlap.

Answer: C. Methane has one carbon atom bonded to four hydrogen atoms, and each covalent bond is a shared pair made of **one electron from carbon (a dot)** and **one electron from hydrogen (a cross)**. Carbon then has a share of 8 outer electrons and each hydrogen a share of 2. A has the atoms the wrong way round; B shows only two bonds (CH_2); D shows both electrons in every bond coming from carbon, which is impossible because carbon has only 4 outer electrons, and the hydrogen atoms' own electrons are missing.

Question C2.15

The boiling point of a substance relates to the type of bonding present in the substance. Two elements, X and Y, combine to form a liquid with the relatively low boiling point of 120 °C. Which of the alternatives A–D in Table C2.09 correctly describes this substance?

Option	Element X	Element Y	Type of bonding
A	metal	metal	covalent
B	non-metal	non-metal	ionic
C	non-metal	non-metal	covalent
D	metal	non-metal	ionic

Answer: C. A liquid with a low boiling point is a **simple covalent (molecular)** substance, because only weak intermolecular forces have to be overcome to boil it. Covalent bonds form between **non-metal atoms**. A is wrong because metals are held together by metallic bonding, not covalent bonding. B is wrong because two non-metals share electrons rather than forming ions. D describes an ionic compound, which would be a solid with high melting and boiling points.

Question C2.16

Draw dot-and-cross diagrams of the ionic bonding in the following compounds and give their formula:

(a) lithium fluoride

(b) sodium bromide

(c) potassium chloride

Answer: In each diagram, draw the metal's electrons as crosses and the non-metal's electrons as dots. First draw the two atoms, with an arrow showing the metal's single outer electron moving to the outer shell of the non-metal atom. Then draw each ion in square brackets with its charge at the top right.

(a) Lithium fluoride, **LiF**. The lithium atom (2,1) transfers its one outer electron to the fluorine atom (2,7).

Ion	Electrons in each shell	How it is drawn	Charge
Lithium ion, Li^+	2	The outer shell is now empty; only the full first shell of 2 crosses remains	1+
Fluoride ion, F^-	2,8	Outer shell of 8: 7 dots and 1 cross (the electron from lithium)	1−

(b) Sodium bromide, **NaBr**. The sodium atom (2,8,1) transfers its one outer electron to the bromine atom (2,8,18,7).

Ion	Electrons in each shell	How it is drawn	Charge
Sodium ion, Na^+	2,8	The third shell is now empty; the full second shell of 8 crosses is the outer shell	1+
Bromide ion, Br^-	2,8,18,8	Outer shell of 8: 7 dots and 1 cross (the electron from sodium)	1−

(c) Potassium chloride, **KCl**. The potassium atom (2,8,8,1) transfers its one outer electron to the chlorine atom (2,8,7).

Ion	Electrons in each shell	How it is drawn	Charge
Potassium ion, K^+	2,8,8	The fourth shell is now empty; the full third shell of 8 crosses is the outer shell	1+
Chloride ion, Cl^-	2,8,8	Outer shell of 8: 7 dots and 1 cross (the electron from potassium)	1−

In each compound one electron is transferred, so the ions have charges of 1+ and 1− and combine in a **1 : 1 ratio**. Every ion has a noble gas configuration: Li^+ like helium; F^- and Na^+ like neon; K^+ and Cl^- like argon; Br^- like krypton.

Question C2.17

Figure C2.30 shows the structure of a molecule of methanoic acid. The carbon atom is bonded to a hydrogen atom (C–H), to one oxygen atom by a double bond (C=O) and to a second oxygen atom by a single bond; this second oxygen atom is also bonded to a hydrogen atom (O–H). The shaded region is the single bond between the carbon atom and the oxygen atom of the O–H group. Table C2.10 lists four descriptions of the bonds formed between the carbon and oxygen atoms in the shaded region of the structure. Which of the descriptions A–D is correct?

Option	Carbon	Oxygen
A	shares 1 electron with hydrogen and 3 electrons with oxygen	shares 1 electron with hydrogen and 1 electron with carbon
B	shares 2 electrons with hydrogen and 2 electrons with oxygen	shares 1 electron with hydrogen and 2 electrons with carbon
C	shares 1 electron with hydrogen and 3 electrons with oxygen	shares 1 electron with hydrogen and 2 electrons with carbon

Option	Carbon	Oxygen
D	shares 2 electrons with hydrogen and 2 electrons with oxygen	shares 1 electron with hydrogen and 1 electron with carbon

Answer: A. Carbon has 4 outer electrons and forms four bonds: it shares 1 electron with the hydrogen atom (C–H), 2 electrons with the double-bonded oxygen (C=O) and 1 electron with the oxygen in the shaded region (C–O). That is 1 electron with hydrogen and 3 electrons with oxygen. The shaded oxygen has 6 outer electrons and forms two single bonds: it shares 1 electron with hydrogen (O–H) and 1 electron with carbon (C–O), keeping its other 4 electrons as two non-bonding pairs. B and D are wrong because only one hydrogen atom is bonded to the carbon; C is wrong because the shaded C–O bond is a single bond, so the oxygen shares only 1 electron with carbon.

Activity C2.02 — Question 1

Figure C2.31 shows part of a mind map on atoms, ions and molecules. A central cloud, 'Atoms, ions and molecules', has three branches, but only the lower branch has been completed. It leads to a cloud labelled 'Atomic theory' (based on Dalton's idea), surrounded by five boxes:

- Atoms are the smallest particles that take part in a …… reaction.
- Atoms of different elements can combine to make the molecules of a ……
- The atoms of the different elements differ in size.
- A pure element contains only atoms with the same number of …… in the nucleus.
- Atoms of an element are each given their own symbol.

Copy the current map and complete the 'atomic theory' group by filling in the missing words.

Answer:

- Atoms are the smallest particles that take part in a **chemical** reaction.
- Atoms of different elements can combine to make the molecules of a **compound**.
- A pure element contains only atoms with the same number of **protons** in the nucleus.

Activity C2.02 — Question 2

Draw new clouds on the unoccupied branches: the upper branch labelled 'ionic bonding' and the middle branch labelled 'covalent bonding'.

Answer: Draw a cloud at the end of the upper branch containing the words **Ionic bonding**, and a cloud at the end of the middle branch containing the words **Covalent bonding**. Each new cloud becomes the centre of a group of ideas, in the same way as the 'Atomic theory' cloud.

Activity C2.02 — Question 3

Add boxes around each new cloud giving the key details of that type of bonding. Include as many ideas as you can.

Answer: Boxes around the **Ionic bonding** cloud:

- Formed between a metal and a non-metal.
- Electrons are transferred from the metal atoms to the non-metal atoms.
- Metal atoms lose electrons to form positive ions (cations), e.g. Na^+, Mg^{2+}.
- Non-metal atoms gain electrons to form negative ions (anions), e.g. Cl^-, O^{2-}.
- The ions formed have the electronic configuration of a noble gas.
- The ionic bond is the strong electrostatic attraction between oppositely charged ions.
- The ions form a giant ionic lattice, so ionic compounds have high melting and boiling points.
- They conduct electricity when molten or dissolved in water, and are often soluble in water.
- Examples: NaCl, MgO, $CaCl_2$.

Boxes around the **Covalent bonding** cloud:

- Formed between non-metal atoms.
- A covalent bond is a shared pair of electrons, usually one electron from each atom.
- Each atom gains a share of 8 outer electrons (2 for hydrogen), a noble gas configuration.
- There are single, double and triple bonds: Cl–Cl, O=O, N≡N.
- Simple molecules include H_2, Cl_2, HCl, H_2O, NH_3, CH_4 and CO_2.
- Weak intermolecular forces give low melting and boiling points; many are liquids or gases.
- Simple covalent substances do not conduct electricity.
- The bonding is shown using dot-and-cross diagrams of the outer electrons.

Activity C2.02 — Question 4

Using another coloured pen, can you make a new link between one of the lower boxes and the middle cloud?

Answer: Yes. For example, draw a coloured line from the box 'Atoms of different elements can combine to make the molecules of a compound' to the **Covalent bonding** cloud, and label it: the atoms in molecules of compounds such as water, ammonia and methane are held together by covalent bonds (shared pairs of electrons). Another good link is from 'A pure element contains only atoms with the same number of protons' to Covalent bonding: non-metal elements such as H_2, O_2, N_2 and Cl_2 are molecules made of identical atoms joined by covalent bonds.

Reflection — Diagrams and the Links Between Types of Bonding

This chapter involves drawing different diagrams and visual representations of molecules and structures. Do you find drawing these diagrams useful? Do these visualisations help you to understand the topic? Could you use diagrams in other topics in chemistry to help develop your understanding? Diagrams can also be used to organise information, as in Figure C2.36, a flow chart summarising bonding: elements are either metals (metallic bonding, giving giant metallic lattices) or non-metals (covalent bonding, giving simple molecules or giant molecular lattices; the noble gases exist as separate atoms); compounds are either metal + non-metal(s) (ionic bonding, giving giant ionic lattices) or non-metal + non-metal(s) (covalent bonding, giving simple molecules or giant molecular lattices). How does this help you understand the links between the types of bonding we have discussed?

Answer: A model answer: Drawing dot-and-cross diagrams makes it clear which electrons are transferred or shared, and drawing structures such as lattices and layers helps to explain properties like melting point and conductivity. Diagrams would also help in other topics, e.g. particle diagrams for states of matter and apparatus diagrams for experiments. Figure C2.36 shows in one picture that the **types of element involved decide the bonding** (metals alone give metallic bonding, a metal with a non-metal gives ionic bonding, and non-metals alone give covalent bonding) and that each type of bonding leads to a particular type of structure. It also shows that covalent bonding is found in both elements and compounds, and that the noble gases are the only elements with no bonding.

Question C2.18

Which one of the following elements exists as a giant covalent structure?

A carbon

B iodine

C helium

D oxygen

Answer: A (carbon). Carbon exists as diamond and graphite, giant covalent structures in which every atom is joined to its neighbours by strong covalent bonds. Iodine (I_2) and oxygen (O_2) are made of simple molecules, and helium exists as separate atoms.

Question C2.19

Metals such as copper are bonded together by the attraction between metal ions and a 'sea' of delocalised electrons surrounding them. Which of the properties of metals (A–C) is not explained by this type of bonding?

A electrical conductivity

B malleability

C reaction with acids

Answer: C (reaction with acids). This is a chemical property, which depends on how reactive the metal is, not on the model of metallic bonding. Electrical conductivity is explained by the mobile delocalised electrons moving through the structure, and malleability is explained by the layers of positive ions sliding over each other while the attraction to the sea of electrons, acting in all directions, holds the structure together.

Activity C2.03 — Extending the Mind Map

The types of bonding and particles described in the mind map you devised in Activity C2.02 (Part 1) produce several different giant structures present in solid substances. Look back at the mind map you created and see how you can extend it to include these structures.

Answer: Add a new cloud, **Giant structures**, linked to the bonding clouds, with boxes such as:

- **Giant ionic lattice** (linked to Ionic bonding) — a regular arrangement of alternating positive and negative ions held together by strong electrostatic forces, e.g. sodium chloride; high melting points; conducts when molten or dissolved.
- **Giant covalent structures** (linked to Covalent bonding) — diamond (each carbon bonded to 4 others; very hard; does not conduct), graphite (layers of hexagons, each carbon bonded to 3 others; soft, slippery and conducts) and silica, SiO_2; all have very high melting points.
- **Giant metallic lattice** (linked to a new Metallic bonding cloud) — positive metal ions in a 'sea' of delocalised electrons; metals conduct electricity and are malleable.

Also add a box contrasting these with **simple molecules**, which have weak intermolecular forces and so low melting points, and use a different colour to link each structure to its type of bonding.

Project C2.01 — Modelling the Structure of an Atom

Images of atoms often show electrons orbiting like planets (Figure C2.37a: a planetary model of an atom with three electrons orbiting a nucleus). Working in groups, construct a 3D model of either a planetary model of a simple atom, or a model showing the energy levels that electrons can occupy in an atom, representing the energy levels like trays with a certain number of electrons on each tray (Figure C2.37b: discs of laminated card threaded on a skewer and separated by short pieces of plastic straw, with a styrofoam ball). Having constructed your model, take a moment to assess how well it represents an atom. Be critical of your model and discuss its limitations with your group. How well does your model represent what you have learnt in this chapter?

Answer: A model answer for a planetary model of lithium, the atom shown in Figure C2.37a:

- Nucleus: 3 protons and 4 neutrons (lithium-7), made from small balls of two colours stuck together in the centre.
- Electrons: 3 small balls on wire rings, 2 on the inner ring (first shell) and 1 on the outer ring (second shell), giving 2,1.

What the model shows well:

- The protons and neutrons are together in a nucleus at the centre, with the electrons outside it, in shells.
- The number of electrons equals the number of protons, so the atom is neutral.
- The shells fill from the inside, and the first shell holds only 2 electrons (in a tray model: 2 on the first tray, then up to 8 on each of the next trays).

Limitations:

- It is not to scale: the real nucleus is tiny compared with the whole atom, which is mostly empty space, and an electron has almost no mass (1/1840 of a proton) but is shown as a ball of similar size.
- Electrons do not really move like planets on flat, fixed tracks, and the model cannot show their movement.
- It cannot show the charges, or the electrostatic attraction that holds the electrons to the nucleus.
- A tray model shows the energy levels and how many electrons each holds, but not the distances from the nucleus or the shape of the atom.

Overall, the model represents the arrangement of the subatomic particles and the electronic configuration well, but not the sizes, masses, charges or forces involved.

Practice Question 1

The structure of an atom is defined by two numbers: the proton (atomic) number and the mass (nucleon) number. Identify the electronic configuration of an atom with proton number 5 and mass number 11. (1 mark)

A 2,8,1

B 3,2

C 2,3

D 1,8,2

Answer: C (2,3). Proton number 5 means 5 protons, so the neutral atom has 5 electrons: 2 fill the first shell and the remaining 3 go into the second shell. The mass number (11) does not affect the electron arrangement. A has 11 electrons (it confuses the mass number with the number of electrons), and B and D break the rule that the first shell holds 2 electrons and fills first. [1]

Practice Question 2

Elements in the Periodic Table are arranged in order of their proton number. (10 marks)

(a) Define proton number. [1]

(b) Argon, proton number 18, has a larger mass number than potassium, proton number 19. Explain why this is, considering the mass and atomic numbers. [2]

(c) Copy and complete this table to give the properties of the three particles that make up atoms. [5]

Particle	Charge	Mass	Position in atom
Proton	+	?	?
Neutron	?	1	In the nucleus
Electron	−	?	?

(d) Atom X has only two of the three particles listed in the table above. State which element Atom X is an atom of. [1]

(e) Argon and helium are both noble gases. Give one way in which their electron structures are similar. [1]

Answer:

(a) The proton number is the **number of protons in the nucleus** of an atom. [1]

(b) The mass number is the total number of **protons + neutrons** in the nucleus. Although an argon atom has one proton fewer than a potassium atom, it has **more neutrons**: the commonest argon atom, argon-40, has 18 protons and 22 neutrons, while the commonest potassium atom, potassium-39, has 19 protons and 20 neutrons. The Periodic Table is arranged by proton number, not by mass number. [2]

(c) One mark for each correct missing entry: [5]

Particle	Charge	Mass	Position in atom
Proton	+1	1	In the nucleus
Neutron	0 (no charge)	1	In the nucleus
Electron	−1	1/1840 (negligible)	In shells (energy levels) around the nucleus

(d) **Hydrogen**. The commonest hydrogen atom has one proton and one electron but **no neutrons**. [1]

(e) Both have a **full outer shell** of electrons: helium has 2 electrons in its only shell, and argon has 8 electrons in its outer shell (2,8,8). [1]

Practice Question 3

$^{12}_{6}C$ and $^{14}_{6}C$ are two isotopes of the element carbon. (6 marks)

(a) Describe how the structure of their atoms differs. [3]

(b) Carbon reacts with oxygen to produce carbon dioxide. Give the electronic configuration of an oxygen atom. [1]

(c) Germanium (proton number 32) is an element in the same group as carbon.

(c)(i) State how many electrons germanium has in its outer electron shell. [1]

(c)(ii) State which period of the Periodic Table germanium is found in. [1]

Answer:

(a) Both atoms have the **same number of protons (6)** and the **same number of electrons (6)**, arranged 2,4. They differ in the number of **neutrons**: carbon-12 has **6 neutrons** (12 − 6) and carbon-14 has **8 neutrons** (14 − 6), so a carbon-14 atom has two more neutrons and a greater mass. [3]

(b) Oxygen has 8 electrons, arranged **2,6**. [1]

(c)(i) **4**. Germanium is in the same group as carbon (Group IV), and the group number equals the number of outer electrons. [1]

(c)(ii) **Period 4**. Its 32 electrons are arranged 2,8,18,4, so it has four occupied shells, and the period number equals the number of shells. [1]

Practice Question 4

The element copper has two main isotopes, $^{63}_{29}Cu$ and $^{65}_{29}Cu$. Both isotopes have the same chemical properties. Explain why the chemical properties of these two isotopes are the same. (2 marks)

Answer: Both isotopes have **29 protons** and therefore **29 electrons**, so they have the **same electronic configuration**, with the same number of outer electrons. Chemical reactions involve the **electrons** (especially the outer electrons), so both isotopes react in exactly the same way. They differ only in their number of neutrons (34 and 36), which does not affect chemical reactions. [2]

Practice Question 5

There are various different types of structure that compounds can form, depending on the nature of their bonding. A substance has an ionic structure that can be represented as shown in the figure: a regular grid of circles in three rows of four, in which large circles marked − alternate with small circles marked +, so that every ion is next to ions of the opposite charge (top row − + − +, middle row + − + −, bottom row − + − +). (3 marks)

(a) Suggest what this substance could be. [1]

A iodine

B water

C potassium bromide

D diamond

(b) Give reasons for your answer to part (a). [2]

Answer:

(a) **C (potassium bromide)**. [1]

(b) The figure shows a regular lattice of **alternating positive and negative ions**, each surrounded by ions of the opposite charge: a giant ionic lattice. Potassium bromide is the only substance listed that is formed from a **metal and a non-metal**: potassium atoms lose an electron to form K^+ ions (the small circles) and bromine atoms gain an electron to form Br^- ions (the large circles). Iodine and water are made of simple covalent molecules and diamond is a giant covalent structure, so none of them contains ions. [2]

Practice Question 6

When potassium reacts with chlorine the ionic compound potassium chloride is formed. The electronic configurations of the two ions formed are the same. The figure shows the two ions, each in square brackets: a potassium ion with shells of 2, 8 and 8 electrons, all drawn as crosses, with a + charge; and a chloride ion with shells of 2, 8 and 8 electrons drawn as dots, except for one cross in the outer shell, with a − charge. (6 marks)

(a) Explain what has happened during the reaction to form these two ions. [2]

(b) When two chlorine atoms react with each other ions are not formed. Sketch a dot-and-cross diagram to show the bond between two chlorine atoms. Show outer shell electrons only. [2]

(c) Give the name of this type of bond. [1]

(d) Give a difference in physical properties between chlorine and potassium chloride. [1]

Answer:

(a) The potassium atom (2,8,8,1) has **lost its one outer electron**, which has been **transferred to the chlorine atom** (2,8,7). The potassium atom becomes a positive potassium ion, **K^+**, and the chlorine atom becomes a negative chloride ion, **Cl^-**. Both ions now have the configuration **2,8,8**, a full outer shell like the noble gas argon. [2]

(b) Draw two overlapping circles (the outer shells), each with Cl in the centre:

- In the overlap: **one shared pair** of electrons, one dot from the first chlorine atom and one cross from the second.
- Outside the overlap: **three more pairs of dots** (6 electrons) on the first chlorine atom and **three more pairs of crosses** (6 electrons) on the second.
- Each chlorine atom now has 8 electrons in its outer shell, counting the shared pair. [2]

(c) A (single) **covalent bond**. [1]

(d) Chlorine is a **gas** at room temperature with **low** melting and boiling points (simple molecules with weak intermolecular forces), whereas potassium chloride is a **solid** with **high** melting and boiling points (a giant ionic lattice with strong electrostatic forces). Another correct difference: potassium chloride conducts electricity when molten or dissolved in water, but chlorine does not conduct. [1]

Practice Question 7

Carbon exists in two forms, diamond and graphite. Both forms have very high melting points. Graphite conducts electricity but diamond does not. (4 marks)

(a) Explain why graphite has a high melting point and conducts electricity. [2]

(b) Graphite is soft and can be used as a lubricant, but diamond is very hard. Explain this difference. [2]

Answer:

(a) Graphite has a **giant covalent structure** in which each carbon atom is joined to three others by **strong covalent bonds**; a lot of energy is needed to break these many bonds, so the melting point is

high. Each carbon atom uses only three of its four outer electrons in bonding, so there are **free (delocalised) electrons** that can move between the layers carrying charge, and graphite conducts electricity. [2]

(b) In graphite the atoms are arranged in **layers** held to each other only by **weak forces**, so the layers can **slide over each other** easily, making graphite soft and slippery. In diamond each carbon atom is bonded to **four** others by strong covalent bonds in a rigid **three-dimensional (tetrahedral)** network that extends in all directions, so nothing can slide and diamond is very hard. [2]

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12. Exam Focus Priorities (Ranked by Frequency)

1. **Counting protons, neutrons and electrons** — using proton number and mass number (neutrons = A − Z), including ions such as Na^+ and Cl^-, and recalling the relative masses and charges of the three subatomic particles.
2. **Electronic configurations** — writing configurations for the first 20 elements (shells of 2, 8, 8) and linking group number to outer electrons and period number to the number of shells.
3. **Ionic bonding and dot-and-cross diagrams** — electron transfer from metal to non-metal, ion charges, square brackets, and working out formulae such as NaCl, MgO and $CaCl_2$.
4. **Covalent bonding and dot-and-cross diagrams** — shared pairs in H_2, Cl_2, HCl, H_2O, NH_3 and CH_4, and the double and triple bonds in O_2, N_2, CO_2 and C_2H_4.
5. **Properties linked to structure and bonding** — explaining the melting points, boiling points and electrical conductivity of ionic and simple molecular substances (strong electrostatic forces versus weak intermolecular forces).
6. **Diamond and graphite** — comparing their structures (4 or 3 bonds per carbon atom, layers) and using them to explain hardness, lubrication, electrical conductivity and uses.
7. **Isotopes** — the definition, working out the neutrons in isotopes such as chlorine-35 and chlorine-37, and explaining why isotopes have the same chemical properties.
8. **Metallic bonding** — describing positive ions in a sea of delocalised electrons and using it to explain electrical conductivity and malleability.
9. **Elements, compounds and mixtures** — the definitions and the differences between compounds and mixtures, including the iron and sulfur example.
10. **Noble gases and stability** — full outer shells explain why the noble gases are unreactive and why other atoms form ions or share electrons.

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13. Quick Revision Card

- **Element:** one type of atom (same proton number); **compound:** elements chemically combined in fixed proportions; **mixture:** not chemically bonded and separated by physical methods.
- **Proton:** mass 1, charge +1, in the nucleus. **Neutron:** mass 1, charge 0, in the nucleus. **Electron:** mass 1/1840, charge −1, in shells.
- **Z = number of protons = number of electrons** (in an atom); **A = protons + neutrons**; **neutrons = A − Z**.
- **Shells hold 2, 8, 8:** e.g. Na 2,8,1; Cl 2,8,7; Ca 2,8,8,2.
- **Group number = outer electrons; period number = number of shells.**
- **Noble gases:** full outer shells, very stable and unreactive; other atoms bond to reach this configuration.
- **Isotopes:** same proton number, different numbers of neutrons; same chemical properties because they have the same electronic configuration.
- **Ionic bonding (metal + non-metal):** electrons transferred; the metal forms positive ions and the non-metal negative ions, held by strong electrostatic attraction.

- **Ionic compounds:** giant lattice; high melting and boiling points; often soluble in water; conduct only when molten or dissolved.
- **Covalent bond (non-metals):** a shared pair of electrons; single (Cl–Cl), double (O=O, O=C=O) and triple (N≡N) bonds.
- **Simple molecules:** strong covalent bonds within, weak intermolecular forces between, so low melting and boiling points; do not conduct.
- **Diamond:** 4 bonds per carbon, tetrahedral; very hard, very high melting point, does not conduct; cutting tools and jewellery.
- **Graphite:** 3 bonds per carbon in layers that slide, with free electrons; soft, used as a lubricant and in pencils, conducts (electrodes).
- **Metallic bonding:** positive ions in a 'sea' of delocalised electrons; metals conduct electricity and are malleable.